

Shaswata Mitra

Ph.D. Candidate · Department of Computer Science
The University of Alabama, Tuscaloosa, AL

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RESEARCH PROFILE

Ph.D. candidate in **Computer Science** at **The University of Alabama**, with core research interests in **generative and agentic AI for autonomous cybersecurity**, automating threat-intelligence contextualization and secure agentic integration across security pipelines, and **adversarial AI for explainability**, using latent regularization and targeted adaptation to make models robust against adversarial manipulation. Built first-author frameworks including *LocalIntel* (FPS 2024, **Best Paper Award**), with complementary industry experience in auditable agentic workflow automation and context-aware retrieval at **Fujitsu Research**. Author of **20+ publications** across IEEE BigData, IJCNN, ICMLA, CAI, CIC, IEEE Access, and ACM Transactions, with **2 patent filings**, **h-index of 12**, **i10-index of 15**, and **495+** citations.

EDUCATION

The University of Alabama <i>Ph.D. in Computer Science (Transferred from MSU); GPA: 4.0/4.0</i> - Generative AI, Agentic Systems, Cybersecurity, Representation Learning - Advised by: Dr. Sudip Mittal	Aug. 2025 – Present Tuscaloosa, AL, USA
Mississippi State University <i>M.S. in Computer Science; GPA: 4.0/4.0</i> - AI, NLP, Cybersecurity, Knowledge Graph - Advised by: Dr. Sudip Mittal	Aug. 2021 – Jul. 2025 Starkville, MS, USA
West Bengal University of Technology <i>B.Tech in Computer Science and Engineering</i>	Aug. 2015 – May 2019 Kolkata, WB, India

EXPERIENCE

Fujitsu Research of America <i>Research Intern</i> - Conducting research on agentic coding framework for data privacy, with anticipated patent filing. - Developed an auditable multi-agent workflow automation using graph ML, resulting in a patent filing. - Designed a context-aware data indexing and retrieval model, resulting in a patent filing.	May 2025 – Aug. 2025; May. 2026 – Nov. 2026 San Jose, CA, USA
The University of Alabama <i>Graduate Research Assistant</i> - Developed a embedding model to capture semantic relation from heterogeneous sources. - Developed benchmark to assess domain-alignment, privacy, reasoning, and explainability of LLMs. - Built grounded CTI generation utilizing LLM and diffusion models to mitigate novel threats.	Aug. 2025 – May. 2026 Tuscaloosa, AL, USA
National Strategic Planning & Analysis Research Center (nSPARC) <i>Research Intern</i> - Conducted research on software vulnerability classification using Neuro-Symbolic AI. - Built a co-relation quantifier to map entity and relation extraction from heterogeneous reports.	May. 2024 – Aug. 2024 Starkville, MS, USA
Mississippi State University <i>Graduate Research Assistant</i> - Investigated polymorphic malware detection through control flow graph (CFG) utilizing ML. - Build residual vision model for off-road traversability estimation from aerial images. - Designed adaptive anomaly quantifier using clustering approach over network data for DDoS mitigation. - Investigated GNN to improve knowledge representation for different classifiers in cyber operations.	Aug. 2021 – May 2025 Starkville, MS, USA
Tata Consultancy Services (TCS) <i>Software Engineer - Full-stack Developer</i> - Created a personalized customer engagement analysis pipeline for assistive marketing. - Served as project interviewer to recruit and mentor new employees to be oriented with project code.	July 2019 – July 2021 Kolkata, WB, India

Journal Articles

Neupane, S., **Mitra, S.**, Mittal, S., Gaur, M., Golilarz, N. A., Rahimi, S., and Amirlatifi, A. “MedInsight: A Multi-Source Context Augmentation Framework for Generating Patient-Centric Medical Responses using Large Language Models.” *ACM Transactions on Computing for Healthcare*, 6(2), 1–19, 2025.

Neupane, S., **Mitra, S.**, Fernandez, I. A., Saha, S., Mittal, S., Chen, J., Pillai, N., and Rahimi, S. “Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities.” *IEEE Access*, 2024.

Chakraborty, T., **Mitra, S.**, Mittal, S., and Young, M. “AIAdaptive.POW: An AI-Assisted Proof of Work (POW) Framework for DDoS Defense.” *Software Impacts (Elsevier)*, 13, 100335, 2022.

Conference and Workshop Proceedings

Mitra, S., Neupane, S., Duclos, M., Mittal, S., Piplai, A., Rahman, M. R., Ziegler, E., and Rahimi, S. “FALCON: Autonomous Cyber Threat Intelligence Mining with LLMs for IDS Rule Generation.” *InProceedings, IEEE International Conference on Collaboration and Internet Computing (CIC)*, 2026. (Accepted)

Mitra, S., Neupane, S., Chakraborty, T., Tripathi, H., Mittal, S., Rahimi, S., and Piplai, A. “Find Before You Fine-Tune: A Diagnostic Study of Small LLMs for Cybersecurity QA.” *InProceedings, IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026. (Accepted)

Patel, R., **Mitra, S.**, Guida, M., Iannucci, S., Mittal, S., and Rahimi, S. “Agentra: A Supervisable Multi-Agent Framework for Enterprise Intrusion Response.” *InProceedings, IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026. (Accepted)

Ryan, A., Noor, S. S., Erfan, M., **Mitra, S.**, Mittal, S., and Rahman, M. R. “Evaluating Open-Source LLMs for Multi-Label ATT&CK Technique Classification on CTI Reports.” *InProceedings, IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2026. (Accepted)

Panigrahi, D., Patel, R., **Mitra, S.**, Mittal, S., and Rahimi, S. “IRSDA: An Agent-Orchestrated Framework for Enterprise Intrusion Response.” *IEEE International Conference on Big Data (BigData)*, 2025.

Sanchez, I., **Mitra, S.**, Piplai, A., and Mittal, S. “Semantic-Aware Contrastive Fine-Tuning: Boosting Multimodal Malware Classification with Discriminative Embeddings.” *InProceedings, International Joint Conference on Neural Networks (IJCNN)*, 2025.

Neupane, S., **Mitra, S.**, Fernandez, I. A., Mittal, S., and Rahimi, S. “TwinExplainer: Explaining Predictions of an Automotive Digital Twin.” *InProceedings, IEEE Conference on Artificial Intelligence (CAI)*, 1070–1077, 2025.

Mitra, S., Neupane, S., Chakraborty, T., Mittal, S., Piplai, A., Gaur, M., and Rahimi, S. “LocalIntel: Generating Organizational Threat Intelligence from Global and Local Cyber Knowledge.” *InProceedings, International Symposium on Foundations & Practice of Security (FPS) — Best Paper Award*, 2024.

Panigrahi, D., **Mitra, S.**, Neupane, S., Mittal, S., and Blakely, B. A. “IRSKG: Unified Intrusion Response System Knowledge Graph Ontology for Cyber Defense.” *IEEE International Conference on Big Data (BigData)*, 2024.

Neupane, S., Tripathi, H., **Mitra, S.**, Bozorgzad, S., Mittal, S., Rahimi, S., and Amirlatifi, A. “ClinicSum: Utilizing Language Models for Generating Clinical Summaries from Patient-Doctor Conversations.” *IEEE International Conference on Big Data (BigData)*, 2024.

Fernandez, I. A., Neupane, S., Chakraborty, T., **Mitra, S.**, Mittal, S., Pillai, N., Chen, J., and Rahimi, S. “A Survey on Privacy Attacks Against Digital Twin Systems in AI-Robotics.” *InProceedings, IEEE International Conference on Collaboration and Internet Computing (CIC)*, 70–79, 2024.

Rosas, S. R., Neupane, S., **Mitra, S.**, and Mittal, S. “ReVisE: Emulated Visual Outfit Generation from User Reviews Using Generative-AI.” *InProceedings, International Conference on Software Engineering and Data Engineering (SEDE)*, Springer, 168–178, 2024.

Al Khatib, H. S., Neupane, S., **Mitra, S.**, Tripathi, H., Mittal, S., Rahimi, S., Marhamati, N., and Bozorgzad, S. “Large Language Models for Engineering Patient Journey Knowledge Graphs.” *InProceedings, Association for the Advancement of Artificial Intelligence Fall Symposium (AAAI)*, 2024.

Mitra, S., Torri, S. A., and Mittal, S. “Survey of Malware Analysis through Control Flow Graph using Machine Learning.” *IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2023.

Chakraborty, T., **Mitra, S.**, and Mittal, S. “CAPoW: Context-Aware AI-Assisted Proof of Work based DDoS Defense.” *InProceedings, International Conference on Security and Cryptography (SECRYPT)*, 62–72, 2023.

Chakraborty, T., **Mitra, S.**, Mittal, S., and Young, M. “A Policy Driven AI-Assisted PoW Framework.” *InProceedings, IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2022.

Mitra, S., Piplai, A., Mittal, S., and Joshi, A. “Combating Fake Cyber Threat Intelligence using Provenance in Cyber-security Knowledge Graphs.” *IEEE International Conference on Big Data (BigData)*, 2021.

Pre-prints

Dai, Y., **Mitra, S.**, Patel, R., Sun, Y., Chen, S., Gong, J., Mittal, S., Rahimi, S., Jia, X., and Yu, R. “Staying on the Attractor: Supervising Neural Surrogates of 3D Turbulence Where They Leave It.” (*Under Review*), 2026.

Tripathi, H., Neupane, S., **Mitra, S.**, Mittal, S., Golilarz, N. A., and Rahimi, S. “Gated Activation Steering for Reducing Sycophancy & Hallucination in Medical Question Answering.” (*Under Review*), 2026.

Mitra, S., Patel, R., Mittal, S., Rahman, M. R., and Rahimi, S. “Agentic AI for Cyber Defense: A Survey of Failure Modes, Applications, and Research Opportunities.” (*Under Review - Minor Revisions - ACM Transactions*), 2026.

Hoque, M. N., **Mitra, S.**, Neupane, S., Mittal, S., and Rahimi, S. “Evaluating the NIST Bugs Framework Against CWE as a Successor for Automated Vulnerability Classification.” (*Under Review*), 2026.

Mitra, S., Patel, R., Mittal, S., and Rahman, M., and Rahimi, S. “AgenticCyOps: Securing Multi-Agentic AI Integration in Enterprise Cyber Operations.” (*Under Review*) *arXiv:2603.09134*, 2026.

Tamanna, M., **Mitra, S.**, et al. “What Are Adversaries Doing? Automating Tactics, Techniques, and Procedures Extraction: A Systematic Review.” (*Under Review*) *arXiv:2604.02377*, 2026.

Patel, R., **Mitra, S.**, Amebley, D., Akinrele, T., Dibbo, S., and Rahimi, S. “The Value of Spike Timing: A Leakage-Resistant Benchmark of SNN Design Choices for Network Intrusion Detection” (*Under Review*) *arXiv:2606.01442*, 2026.

Panda, A., Panigrahi, D., **Mitra, S.**, Mittal, S., and Rahimi, S. “Transfer Learning Applied to Computer Vision Problems: Survey on Current Progress, Limitations, and Opportunities.” *arXiv:2409.07736*, 2024.

Mitra, S., Chakraborty, T., Neupane, S., Piplai, A., and Mittal, S. “Use of Graph Neural Networks in Aiding Defensive Cyber Operations.” *arXiv:2401.05680*, 2024.

Moore, C., **Mitra, S.**, Pillai, N., Moore, M., Mittal, S., Bethel, C., and Chen, J. “URA*: Uncertainty-Aware Path Planning using Image-Based Aerial-to-Ground Traversability Estimation for Off-Road Environments.” *arXiv:2309.08814*, 2023.

GRANT WRITING EXPERIENCE

Contributor, NSF Proposals – with Dr. Sudip Mittal

- Helped advisor in writing NSF proposals on AI for cybersecurity:
 - * **SaTC-EDU**: Inculcate a Culture of Preparedness against AI Security Threats to Pervasive Robotic Systems (\$399,978)
 - * **Collaborative Research**: Promoting Undergraduate Discovery in Cybersecurity for Multi-Cloud Machine Learning Operations (\$164,107)

TEACHING EXPERIENCE

Guest Lecturer — CS-591 — AI for Cyber Security

Fall 2025

The University of Alabama

AL, USA

- Designed and delivered lectures for the “AI in Cybersecurity” course, covering deep learning models for threat classification, multi-modal learning, language model alignment for knowledge contextualization.
- Developed in-class assignments aligned with lecture content, building assessment tools and delivering course material.

Guest Lecturer — CSE-6643 — AI-Robotics

Spring & Fall 2024

Mississippi State University (NSF-supported)

MS, USA

- Designed and delivered lectures for the “Security Considerations in AI-Robotics Systems” courses, covering AI-robotics design, Human-Computer Interaction (HCI), and security practices.
- Developed in-class assignments aligned with lecture content, building assessment tools and delivering course material.

In-Class Tutorial Mentor — CSE-6990 — AI for Cybersecurity

Fall 2023

Mississippi State University

MS, USA

- Facilitated tool setup and usage, led discussions, provided feedback, and addressed student questions for the “AI for Cybersecurity” course.
- Developed group-work facilitation and mentoring skills through hands-on student support.

MENTORING

2026-Present Michele Guida, Visiting Research Scholar, UA
2026-Present Joseph Boadi, Ph.D. Student (1st Year), UA
2026-Present Shafkat Hassan, Ph.D. Student (1st Year), UA
2026-Present Anita Binoo Joseph, Ph.D. Student (1st Year), UA
2026-Present Md Nazmul Hoque, Ph.D. Student (1st Year), UA
2026-Present Ahmed Ryan, Ph.D. Student (2nd Year), UA
2026-Present Md Erfan, Ph.D. Student (2nd Year), UA
2025-Present Raj Patel, Ph.D. Student (2nd Year), UA
2024-Present Himanshu Tripathi, Ph.D. Student (2nd Year), UA
2024-2025 Azim Bazarov, Graduate (M.S.), MSU → Programmer Analyst, nSPARC
2024-2025 Ivan Montoya Sanchez, Graduate (M.S.), UTEP → Ph.D. Student
2024-2025 Kishore Ginka, (K-12) High School → Undergraduate, University of California, Riverside
2024 Aaryan Panda, (K-12) High School → Undergraduate, University of California, San Diego
2024 Samar Rahimi, (K-12) High School → Undergraduate, University of Southern California

OUTREACH AND PROFESSIONAL DEVELOPMENT

Program Committee

2024–2026 IEEE International Conference on Big Data (BigCyber Workshop)

Session Chair

2026 IEEE International Conference on Machine Learning and Applications — Special Session (Anticipated)
2024 IEEE International Conference on Big Data, (BigCyber) — Special Session

Peer Review

2026 IEEE Transactions on Information Forensics and Security — Reviewed 1 journal article
2026 IEEE ICMLA — Reviewed 3 papers
2025 AAAI — Reviewed 2 papers
2025 ESORICS — Reviewed 1 paper
2024 WITS — Reviewed 1 paper
2024 AAAI — Reviewed 1 paper
2023 IEEE Access — Reviewed 1 journal article
2022 IEEE BigData — Reviewed 2 papers
2022 ACM CODASPY — Reviewed 2 papers
2021 IEEE BigData — Reviewed 3 papers

Invited Talks

2026 Reliable Agentic Workflow Automation using DAG-Based Orchestration — *Fujitsu Research of America*
2025 Agentic AI for Defensive Cyber Operations — *The University of Alabama*
2025 Towards Trustworthy AI: A Neuro-Symbolic Approach — *PATENT Lab*
2025 Context-guided Hierarchical Retrieval for Grounded Reasoning — *Fujitsu Research of America*
2024 Intelligent Administrative Automation for Institutions using LLMs — *nSPARC*
2024 Getting Started with AI: Fundamentals, Tools, and Career Pathways — *AI Club, MSU*
2024 Contextualizing Knowledge from Heterogeneous Data using Generative AI — *Mississippi State University*
2023 Graph Neural Networks for Defensive Cyber Operations — *SECRETS Lab*

AWARDS & RECOGNITION

Graduate Council Fellowship 2026
• Received The University of Alabama Graduate Council Fellowship of \$63,056.
Best Paper Award 2024
• Best Paper at 17th International Symposium on Foundations & Practice of Security (FPS - 2024).

LEADERSHIP & SERVICE

President – AI Club at MSU 2024
• Organized workshops, tutorials, competitions and coordinated with industry leaders for outreach.